

“Babbage has devoted ... years to the **realization of a gigantic idea.**”

Luigi Menabrea, 1842

and divide. It was to be programmed using punch cards held together by string, an idea Babbage copied from the French weaver Joseph Jacquard, who used such punch cards to control the patterns on his looms.

The design for the Analytical Engine had a similar logic structure to that of modern computers. It featured a “store” where intermediate results could be held and a “mill” for arithmetic processing. It could repeat an operation a specified number of times (a process called looping, or iteration) and, depending on the result of an operation, it could choose alternative actions (lateral branching). It could also print results on paper, or as a “stereotype,” impressed on plaster of Paris to form a mold from which a printing plate could be made. Although the Analytical Engine was never built, it represented a crucial development from

the basic arithmetical processing of the Difference Engine and is widely

regarded as the forerunner of the modern computer. Babbage himself did not publish much about the potential of the Analytical Engine, but when his friend Ada Lovelace (see right) translated a paper about it in 1842, Babbage asked her to expand on the original article, “as she understood the machine so well.”

A man of many talents

Babbage was a true polymath. He wrote more than 80 scientific papers on subjects ranging from solar eclipses to geology and statistics; from decimalization to diving bells and lighthouses. He designed a prototype ophthalmoscope (for looking at the retina of the eye) and a “cowcatcher”: a metal frame at the front of a train to clear obstacles from railroad tracks.

Babbage’s ideas about mechanization also went further than his calculating machine. Inspired by the concept of error-free efficiency, he had suggestions to make about human productivity. In 1832, he wrote *On the Economy of Machinery and Manufactures*, which proposed that industrial manufacturing be improved by mechanization. He also suggested a more streamlined division of labor. This idea is still known as the Babbage principle.

